

Math 142, S12

Quiz 7

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Determine whether the sequence $\{a_n\}_{n=1}^{+\infty}$ converges or diverges and find the limit if it converges.

(a) $a_n = \frac{2 + 5n^2}{n + n^2}$

$$\lim_{n \rightarrow \infty} \frac{2 + 5n^2}{n + n^2} = \lim_{n \rightarrow \infty} \frac{\frac{2}{n^2} + 5}{\frac{1}{n} + 1}$$

$$= 5$$

$\Rightarrow$ Convergent

(b) $a_n = n^2 e^{-n}$

$$\begin{aligned} \lim_{n \rightarrow \infty} n^2 e^{-n} &= \lim_{x \rightarrow \infty} x^2 e^{-x} \\ &= \lim_{x \rightarrow \infty} \frac{x^2}{e^x} \\ &= \lim_{x \rightarrow \infty} \frac{2x}{e^x} \\ &= \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0 \end{aligned}$$

$\Rightarrow$ Convergent

Problem 2. (5 points) Determine whether the series is convergent or divergent and find its sum if it is convergent.

(a) $\sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n} \leftarrow \text{Geometric series}$

$$= \sum_{n=1}^{+\infty} \frac{1}{4} \left(\frac{-3}{4} \right)^{n-1}$$

$$r = -\frac{3}{4} \Rightarrow |r| = \frac{3}{4} < 1$$

$\Rightarrow$ Convergent

$$\sum_{n=1}^{+\infty} \frac{(-3)^{n-1}}{4^n} = \frac{a_1}{1-r}$$

$$= \frac{\frac{1}{4}}{1 - (-\frac{3}{4})}$$

$$= \frac{1}{4} \cdot \frac{4}{7}$$

$$= \frac{1}{7}$$

(b) $\sum_{n=1}^{+\infty} \frac{1}{2n}$

$$= \frac{1}{2} \left(\sum_{n=1}^{+\infty} \frac{1}{n} \right) \leftarrow \text{Harmonic series}$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{i} = +\infty \leftarrow \text{from the lecture notes.}$$

$$\Rightarrow 2 \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{i} = +\infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2i} = +\infty$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{1}{2n} \text{ diverges.}$$